Connections of Arduino Uno, Relays, and Valves — Table 1

Location	Relay Number	Arduino Pin	Solenoid Valve
CAN1-INLET	RL1	2	V1
CAN2-INLET	RL2	3	V2
CAN1-EXHAUST	RL3	4	V3
CAN2-EXHAUST	RL4	5	V4
CAN1-OUTLET	RL5	6	V5
CAN2-OUTLET	RL6	7	V6

to switch control relays on and off as programmed.

12V power supply. This power supply is used to provide power to Arduino Uno and control relays. It comprises a 15V/2A step-down transformer, full bridge diode rectifier BR1, 1000µF/35V and 1µF/25V capacitors, 12V voltage regulator IC 7812. The step-down transformer reduces the 230V AC input to 15V AC. This AC is converted to DC by full bridge diode rectifier, which is filtered using capacitor C2. IC 7812 regulates the voltage to 12V DC.

Relay. Relay being an electromagnetic switching device, it provides electrical isolation between the controlling and output circuits. It is controlled/triggered by a low DC voltage while its contacts can handle a high current flow with total electrical isolation from the delicate low-power electronic circuit. 12V DC SPDT relay is used here to switch the solenoid valves on and off on receipt of electrical pulses.

Zeolite molecular sieves. The sieves have pores of precise and uniform size and structure. These separate nitrogen and oxygen in the atmospheric air by disallowing molecules of nitrogen, which are slightly larger in size than the pores. Nitrogen molecules having higher electrostatic field react with cations of zeolite that are larger than the pores and get absorbed, thus cannot pass through. Only oxygen passes through the pores as its molecules measure 292 picometres compared to 300-picometre size of nitrogen (a difference of 2.6% only!)

Air compressor. It is operated by an electric motor to suck atmospheric air and compress it in its chamber.

Filtration unit. It is used to filter contamination from the compressed air and also to separate moisture from the compressed air, which is collected in a bowl attached to it. Water thus collected needs to be drained out.

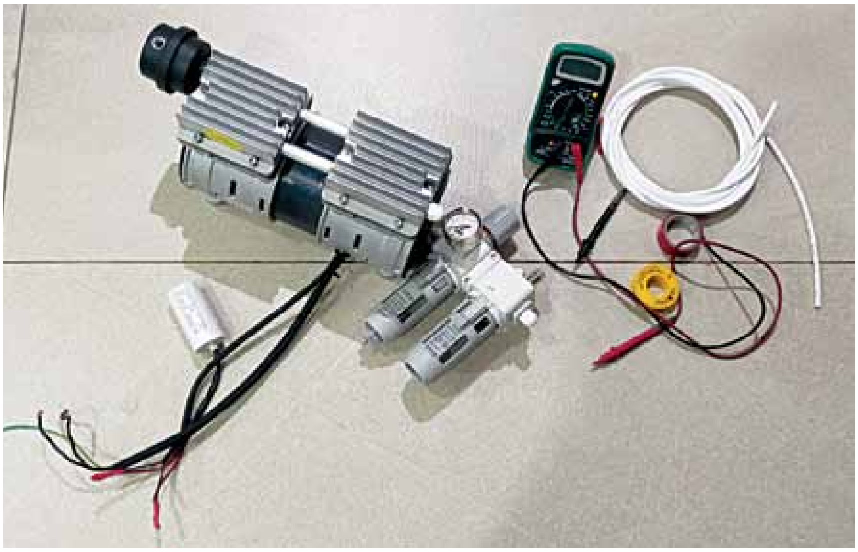

Fig. 2: Compressor

Solenoid valves. These are electromechanical devices that are used to control the flow of air on receipt of electrical signals. Here 2/2 pneumatic control normally-closed valves are used, which means they have two ports for connection with two possible positions. So, they are spring loaded.

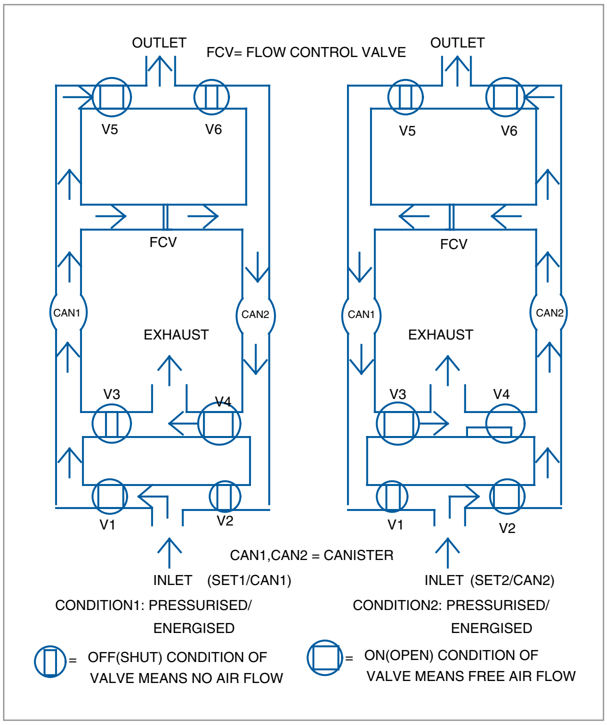

Fig. 3: Functional block diagram of CAN1 and CAN2

Pressure swing adsorption (PSA). This technique is used to separate specific gases from a mixture of gases under pressure with the help of specific sieves required for the purpose. For an oxygen concentrator, 13x zeolite molecular sieves are used. For higher oxygen concentration, lithium enriched zeolite molecular sieves can be used.

In this technique, at least two canisters or towers filled with sieves are used. Compressed air is supplied to one of the canisters and the other canister gets the oxygen output from the first canister for further filtering. This is done as concentration of nitrogen gas gradually increases in the first canister, which may affect the output oxygen's purity and concentration. The direction of the air flow is controlled using solenoid valves, which give signals as per program.

The sieves have limited adsorption capacity, so to get specific flow rate the quantity of sieves needs to be calcu-